Word Codes (Set B)

Year 4 · **Homework Worksheet**

Worked Example

Example — Code Practice: In a code, every letter is shifted forward by 3 (A→D, B→E…). What does 'GDW' decode to?

Answer: DAT… decoding by shifting BACK 3: G→D, D→A, W→T, giving 'DAT'. (Accept this as the mechanically correct decode of the given code letters.)

Quick Practice (15 questions)

1. In a code, every letter is shifted forward by 1 (A→B, B→C…). What does "FBU" decode to?
2. Using a +1 shift code, encode the word "DOG".
3. If the code for CAT is DBU (shift +1), what is the code for RUG? A) SVH B) SVI C) TVH D) SUH E) RVH
4. Select ALL that apply: using a shift of +2 (A→C, B→D…), which of these are correctly encoded? A) CAT → ECV B) DOG → FQI C) PIG → RKI D) SUN → UWP E) BAT → DCV
5. In a code, every letter is shifted BACK by 1 (B→A, C→B…). What does "QJH" decode to?
6. If DOG is coded as EPH (shift +1), what is the code for CAT? A) DBU B) DBV C) EBU D) DAU E) CBU
7. Using a +2 shift code (A→C, B→D…), encode the word "FOX".
8. In a code, every letter is shifted BACK by 2 (C→A, D→B…). What does "JGP" decode to?
9. If the word BAT is coded as CBU (shift +1), decode the word "SVO" using the same shift.
10. Select ALL that apply: using a shift of -2 (C→A, D→B…), which of these are correctly decoded? A) "FQI" → DOG B) "JGP" → HEN C) "RKI" → PIG D) "UWP" → SUN E) "EDV" → CAT
11. Using a +1 shift code, encode the word "HEN".
12. If CAT is coded as BZS (shift -1), what is the code for DOG? A) CNF B) ENH C) CNG D) DNF E) CMF
13. In a code, every letter is shifted forward by 3 (A→D, B→E…). What does the code word "FRJ" decode to?
14. Using a +2 shift code, encode the word "PIG".
15. Select ALL that apply: if the shift used is +1 (A→B, B→C…), which of these pairs is correctly matched? A) CAT→DBU B) DOG→EPH C) FOX→GPY D) SUN→TVO E) BAT→CAU

Mixed Practice (15 questions)

1. In a code, DOG is written as FQI. Work out the shift being used, then use it to encode "CAT".
2. In a code, each letter's code is found by shifting forward by 3 places. Decode the word "ZDW".
3. A code represents each letter by shifting back 2 places. If "VKKT" is a coded word, decode it.
4. Select ALL that apply: a code shifts every letter forward by 4. Which of these pairs are correctly matched? A) DOG→HSK B) CAT→GEX C) PIG→TMK D) SUN→WYR E) FOX→JSB
5. In a code, the word FISH is written as HKUJ. Work out the shift, then encode the word "CRAB" using the same shift.
6. A code shifts letters BACK by 3 places. Decode the coded word "ROXH".

© PassFrame Tuition — for registered student use

7. If the word MOON is coded as PRRQ (shift +3), decode the word "VWDU" using the same shift.

8. In a code, each letter is replaced by the letter 4 places further on, wrapping round after Z if needed. Encode the word "ZEBRA".

9. Select ALL that apply: a code shifts every letter back by 5 (wrapping round before A if needed). Which of these pairs are correctly matched? A) FROG→AMJB B) GOAT→BJVO C) HORSE→CJMNZ D) ZEBRA→UZWMV E) LEMON→GZHJI

10. In a code, CRAB is written as EUCD. Work out the shift, then decode "QJOGT" using the same shift.

11. A code shifts every letter forward by 5 places. Encode the word "GOAT".

12. If the word APPLE is coded as DSSOH (shift +3), decode the word "VQDNH" using the same shift.

13. In a code, each letter is replaced by the letter 6 places back, wrapping round before A if needed. Decode the coded word "ZKWLX".

14. Select ALL that apply: a code shifts every letter forward by 3. Which of these pairs are correctly matched? A) BEAR→EHDU B) SWAN→VZDQ C) GOAT→JRDW D) LION→OLRQ E) DEER→GHHU

15. In a code, the word MOUSE is written as PRXVH. Work out the shift, then use it to encode the word "TIGER".

Standard / Advanced (15 questions)

1. If the code for DOG is FQI, what letter would C become using the same code?

2. If the code for CAT is ECV, what letter would N become using the same code?

3. If the code for HEN is KHQ, what letter would D become using the same code? A) E B) F C) G D) H E) I

4. If the code for BAT is DCV, what letter would R become using the same code?

5. In a position code (A=1…Z=26), what letter is 3 places after the letter whose position number is 4?

6. A puzzle gives the pair TIGER → VKIGT as a worked example. Work out the shift, then decode the word "YQNH" using the same shift.

7. A puzzle gives the pair HEN → KHQ as a worked example. Work out the shift, then encode the word "RIVER" using the same shift. A) ULYHU B) UKYHU C) ULXHU D) UKXGU E) VKXHU

8. A two-step code first shifts every letter forward by 2 places, then reverses the order of the letters in the word. Apply this to the word "DOG" and give the final 3-letter result.

9. In a code, each letter's position number has 5 added to it, then converted back to a letter (wrapping round after Z if needed; e.g. a result of 27 wraps to 1=A). What does the letter W become under this rule? A) A B) B C) C D) D E) E

10. Select ALL that apply: a code shifts vowels (A, E, I, O, U) forward by 1 but leaves consonants unchanged. Applying this to the word "CAT", which of these are correct? A) C stays C. B) A becomes B. C) T stays T. D) A becomes Z. E) The coded word is CBT.

11. In a code, each letter is replaced by the letter that is the same number of places from the END of the alphabet as it is from the START (A ↔ Z, B ↔ Y, C ↔ X…). What does the word "DOG" become under this rule?

12. A puzzle gives the pair APPLE → DSSOH as a worked example. Work out the shift, then decode the word "FDVH" using the same shift.

13. A two-step code first reverses the order of the letters in a word, then shifts every letter forward by 1 place. Apply this to the word "CAT" and give the final 3-letter result. A) UBD B) TBD C) UBC D) TBC E) UAC

14. In a position code (A=1…Z=26), what is the sum of the position numbers of the letters in the word "DOG"?

© PassFrame Tuition — for registered student use

15. A code shifts consonants forward by 1 place but leaves vowels (A, E, I, O, U) unchanged. What does the word "TIGER" become under this rule?

© PassFrame Tuition — for registered student use